

Fitness & Wellness Industry Competitive Market Analysis

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Tools: Python · Pandas · Matplotlib · Seaborn · NumPy

— PROJECT OVERVIEW

This document contains the complete Python source code used to produce the Fitness & Wellness Competitive Market Analysis. The analysis benchmarks five major competitors — Peloton, WHOOP, Lululemon Mirror, Nike Training Club, and Beachbody — across revenue, digital presence, customer sentiment, and strategic positioning.

— DATA SOURCES

Metric	Source
Revenue estimates	Peloton Q4 2023 Earnings Report; Bloomberg Intelligence; Statista
Subscriber counts	Brand investor relations pages; Business of Apps
YoY growth	SEC filings; analyst reports (Morgan Stanley, Citi)
App ratings	Apple App Store & Google Play Store (2024 averages)
Search interest	Google Trends (simulated index for illustration purposes)
Positioning scores	Manually researched — public pricing, reviews, product specs

Note: Revenue figures for private companies (WHOOP, Mirror) are analyst estimates. All data reflects 2023–2024 reporting periods.

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— SECTION 1

Setup & Configuration

Import all required libraries and define global chart settings. The color palette maps consistently to the same brand across every visualization in the analysis.

```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt
import matplotlib.patches as mpatches

import seaborn as sns

import warnings

warnings.filterwarnings("ignore")

# Global style settings – applied to all charts for visual consistency

plt.rcParams.update({
    "figure.facecolor": "white",
    "axes.facecolor": "#F9F9F9",
    "axes.spines.top": False,
    "axes.spines.right": False,
    "font.family": "DejaVu Sans",
    "axes.titlesize": 14,
    "axes.labelsize": 12,
})

# Brand color palette – each color maps consistently to the same brand across all charts

PALETTE = ["#2E7D6B", "#5BAD9B", "#A8D5C8", "#F4A460", "#E07B54"]

BRANDS = ["Peloton", "WHOOP", "Lululemon (Mirror)", "Nike Training Club", "Beachbody"]

print("Setup complete.")
```

— SECTION 2

Brand Overview — Core Business Metrics

Core business metrics compiled from public filings and analyst reports (2023–2024). This DataFrame serves as the foundation for all subsequent analysis.

```
# Core business metrics compiled from public filings and analyst reports (2023-2024)

overview = pd.DataFrame({

    "Brand": ["Peloton", "WHOOP", "Lululemon (Mirror)", "Nike Training Club",
    "Beachbody"],

    "Business_Model": ["Hardware + Subscription", "Subscription + Wearable",
    "Hardware + Subscription", "Freemium App",
    "Subscription + Supplements"],

    "Monthly_Sub_USD": [44, 30, 0, 0, 100],

    "Subscribers_M": [2.96, 1.00, 0.20, 10.00, 0.60],

    "Revenue_2023_M": [717, 200, 120, 550, 230],

    "YoY_Growth_Pct": [-10, 18, -45, 12, -22],

    "App_Rating": [4.8, 4.6, 4.2, 4.7, 4.3],

    "Founded": [2012, 2012, 2016, 2018, 1998]

})

overview
```

— SECTION 3

Revenue & Growth Analysis

Comparing estimated 2023 revenue alongside year-over-year growth reveals a market in transition. The largest brand by revenue (Peloton) is shrinking, while the fastest-growing brand (WHOOP) is still a fraction of its size — signaling an opening for disruption.

```
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Left: Revenue comparison

# Sorted ascending so the longest bar (Peloton) appears at the top

bars = axes[0].barh(
    overview["Brand"], overview["Revenue_2023_M"],
    color=PALETTE, edgecolor="white", height=0.6
)

axes[0].set_xlabel("Estimated Revenue (USD millions)")
axes[0].set_title("2023 Estimated Revenue by Brand", fontweight="bold")

for bar, val in zip(bars, overview["Revenue_2023_M"]):
    axes[0].text(val + 8, bar.get_y() + bar.get_height() / 2,
    f"${val}M", va="center", fontsize=10)
```

```

# Right: YoY growth – red for decline, green for growth
colors_growth = ["#E07B54" if g < 0 else "#2E7D6B" for g in
overview["YoY_Growth_Pct"]]

axes[1].barh(
overview["Brand"], overview["YoY_Growth_Pct"],
color=colors_growth, edgecolor="white", height=0.6
)

axes[1].axvline(0, color="#333333", linewidth=1)
axes[1].set_xlabel("YoY Revenue Growth (%)")
axes[1].set_title("YoY Revenue Growth (2022 to 2023)", fontweight="bold")
for bar, val in zip(axes[1].patches, overview["YoY_Growth_Pct"]):
xpos = val + 0.5 if val >= 0 else val - 0.5
axes[1].text(xpos, bar.get_y() + bar.get_height() / 2,
f"{val:+}%", va="center",
ha="left" if val >= 0 else "right", fontsize=10)

plt.suptitle("Revenue & Growth Snapshot", fontsize=16, fontweight="bold", y=1.02)
plt.tight_layout()
plt.savefig("revenue_growth.png", dpi=150, bbox_inches="tight")
plt.show()

```

Insight: WHOOP leads in growth (+18%) while legacy players Beachbody (-22%) and Mirror (-45%) are in steep decline.

— SECTION 4

Digital Search Interest Trends (2021–2024)

Search interest is a leading indicator of consumer mindshare. This chart uses a simulated Google Trends-style index (0–100) calibrated to known market events such as Peloton's 2022 product recall and executive exit.

```

# Seed for reproducibility – same random state produces identical results each run
np.random.seed(42)

months = pd.date_range("2021-01", periods=36, freq="ME")

# Trend trajectories calibrated to reflect real market events:
# - Peloton: high baseline, gradual decline after 2022 recall
# - WHOOP: low baseline, consistent upward momentum

```

```

# - Mirror: moderate start, steep decline after poor sales reports
# - Nike TC: stable mid-range with slight growth
# - Beachbody: gradual decline from moderate baseline

trends = pd.DataFrame({
    "Month": months,
    "Peloton": np.clip(85 + np.cumsum(np.random.normal(-1.2, 3, 36)), 20, 100),
    "WHOOOP": np.clip(30 + np.cumsum(np.random.normal(1.1, 2, 36)), 10, 90),
    "Mirror": np.clip(40 + np.cumsum(np.random.normal(-1.5, 2, 36)), 5, 80),
    "NikeTC": np.clip(55 + np.cumsum(np.random.normal(0.3, 2, 36)), 30, 100),
    "Beachbody": np.clip(50 + np.cumsum(np.random.normal(-0.8, 2, 36)), 5, 80),
})

fig, ax = plt.subplots(figsize=(13, 5))

labels = ["Peloton", "WHOOOP", "Lululemon (Mirror)", "Nike Training Club", "Beachbody"]
for col, color, label in zip(["Peloton", "WHOOOP", "Mirror", "NikeTC", "Beachbody"],
    PALETTE, labels):
    ax.plot(trends["Month"], trends[col], color=color, linewidth=2.5, label=label)

# Annotate the Peloton recall – a known inflection point in the data
ax.axvline(pd.Timestamp("2022-02"), color="gray", linestyle="--", alpha=0.5)
ax.text(pd.Timestamp("2022-03"), 103, "Peloton recall & CEO exit", fontsize=8,
    color="gray")

ax.set_title("Relative Search Interest – Fitness Brands (2021-2024)",
    fontweight="bold")
ax.set_ylabel("Relative Interest (0-100 Index)")
ax.set_ylim(0, 110)
ax.legend(loc="upper right", fontsize=10)

plt.tight_layout()
plt.savefig("search_trends.png", dpi=150, bbox_inches="tight")
plt.show()

```

Insight: Peloton lost ~30 index points post-recall. WHOOP is the only brand with consistent upward momentum throughout the period.

Competitive Positioning Matrix

Plots brands on two strategically meaningful axes: Price Premium (how expensive is the product) vs. Engagement Depth (how sticky is daily usage). Bubble size reflects estimated market share. Brands in the top-right quadrant command premium pricing AND deep engagement.

```
# Scores manually assigned on a 1-10 scale:

# Price Premium: subscription cost, hardware cost, total cost of ownership
# Engagement Depth: daily active usage rates, session length, churn, community
features

positioning = pd.DataFrame({
    "Brand": ["Peloton", "WHOOP", "Lululemon (Mirror)", "Nike Training Club",
    "Beachbody"],
    "Price_Premium": [8, 6, 7, 2, 5],
    "Engagement_Depth": [9, 8, 6, 5, 7],
    "Market_Share_Pct": [40, 11, 7, 30, 12],
})

fig, ax = plt.subplots(figsize=(9, 7))
ax.set_facecolor("#F5F5F5")

for i, row in positioning.iterrows():
    ax.scatter(
        row["Price_Premium"], row["Engagement_Depth"],
        s=row["Market_Share_Pct"] * 28,
        color=PALETTE[i], alpha=0.85, edgecolors="white", linewidth=1.5, zorder=3
    )
    ax.annotate(
        row["Brand"],
        (row["Price_Premium"], row["Engagement_Depth"]),
        textcoords="offset points", xytext=(12, 4),
        fontsize=10, fontweight="bold", color=PALETTE[i]
    )

# Quadrant dividers at the midpoint of each axis
ax.axhline(5.5, color="gray", linestyle="--", alpha=0.4)
ax.axvline(5.5, color="gray", linestyle="--", alpha=0.4)
```

```

ax.set_xlabel("Price Premium Score (1 = Budget 10 = Ultra Premium)", fontsize=11)
ax.set_ylabel("Engagement Depth Score (1 = Low 10 = Very Sticky)", fontsize=11)
ax.set_title("Competitive Positioning Map\n(Bubble size = estimated market share)")
ax.set_xlim(0, 11)
ax.set_ylim(3, 11)

plt.tight_layout()

plt.savefig("positioning_matrix.png", dpi=150, bbox_inches="tight")

plt.show()

```

Insight: Peloton and WHOOP occupy the premium-engaged quadrant. Nike TC is uniquely accessible at scale but hasn't monetized that reach effectively.

— SECTION 6

Customer Sentiment — App Store Ratings

App store ratings are a proxy for product satisfaction and customer trust. A high percentage of 5-star reviews relative to 1–2 star reviews indicates strong product-market fit and low friction in the user experience.

```

# Rating distributions estimated from App Store and Google Play review breakdowns
brands_clean = ["Peloton", "WHOOP", "Mirror", "Nike TC", "Beachbody"]

# Each row = one brand; columns = % of reviews at each star level (1 to 5)
rating_dist = np.array([
    [3, 2, 4, 12, 79], # Peloton
    [4, 3, 6, 18, 69], # WHOOP
    [7, 5, 9, 20, 59], # Mirror
    [4, 3, 7, 20, 66], # Nike TC
    [8, 6, 10, 22, 54], # Beachbody
])

fig, ax = plt.subplots(figsize=(12, 5))
bottom = np.zeros(len(brands_clean))
star_colors = ["#D32F2F", "#F57C00", "#FBC02D", "#7CB342", "#2E7D32"]

# Build stacked bar from 1-star to 5-star – reveals sentiment skew visually
for j, (star, color) in enumerate(zip([1, 2, 3, 4, 5], star_colors)):
    ax.bar(brands_clean, rating_dist[:, j], bottom=bottom,

```

```

label=f"{star} star", color=color, edgecolor="white", width=0.55)
bottom += rating_dist[:, j]

# Overlay average rating for quick comparison
avg_ratings = [4.8, 4.6, 4.2, 4.7, 4.3]
for i, avg in enumerate(avg_ratings):
    ax.text(i, 103, f"Avg: {avg}", ha="center", fontsize=10, fontweight="bold")

ax.set_ylabel("% of Reviews")
ax.set_title("App Store Rating Distribution by Brand", fontweight="bold", fontsize=14)
ax.set_ylim(0, 115)
ax.legend(loc="lower right", ncol=5, fontsize=9)

plt.tight_layout()
plt.savefig("sentiment.png", dpi=150, bbox_inches="tight")
plt.show()

```

Insight: Peloton leads satisfaction despite business headwinds — product quality remains strong. Beachbody has the weakest sentiment, compounding its revenue decline.

Strategic Scorecard — Radar Chart

Scoring each brand across five strategic dimensions provides a holistic view of competitive strength. No brand leads in all five areas — highlighting where partnerships, acquisitions, or pivots could create advantage.

```
# Five dimensions scored 1-10 based on publicly available information:
# Brand Strength: unaided awareness, NPS, media coverage
# Growth Momentum: YoY revenue, subscriber growth, app downloads
# Product Innovation: R&D investment signals, feature releases, patents
# Community/Retention: churn rates, social engagement, user-generated content
# Monetization: ARPU, LTV, revenue per subscriber

dimensions = ["Brand Strength", "Growth Momentum", "Product Innovation",
              "Community/Retention", "Monetization"]

scores = {
    "Peloton": [9, 4, 7, 9, 7],
    "WHOOP": [6, 9, 9, 7, 8],
    "Mirror": [7, 2, 5, 5, 4],
    "Nike TC": [9, 7, 6, 6, 5],
    "Beachbody": [5, 3, 4, 6, 5],
}

N = len(dimensions)
angles = np.linspace(0, 2 * np.pi, N, endpoint=False).tolist()
angles += angles[:1] # Close the polygon by repeating the first angle

fig, ax = plt.subplots(figsize=(8, 8), subplot_kw=dict(polar=True))
ax.set_facecolor("#F9F9F9")

for (brand, vals), color in zip(scores.items(), PALETTE):
    v = vals + vals[:1]
    ax.plot(angles, v, color=color, linewidth=2, label=brand)
    ax.fill(angles, v, color=color, alpha=0.1)

ax.set_xticks(angles[:-1])
ax.set_xticklabels(dimensions, fontsize=11)
ax.set_ylim(0, 10)
ax.set_yticks([2, 4, 6, 8, 10])
```

```

ax.set_yticklabels(["2", "4", "6", "8", "10"], fontsize=8, color="gray")

ax.set_title("Strategic Scorecard – 5-Dimension Comparison", fontsize=14,
fontweight="bold")

ax.legend(loc="upper right", bbox_to_anchor=(1.35, 1.1), fontsize=10)

plt.tight_layout()

plt.savefig("scorecard.png", dpi=150, bbox_inches="tight")

plt.show()

```

Insight: WHOOP leads on growth and innovation. Peloton leads on community. Nike TC has brand strength but trails on monetization — a significant untapped opportunity.

— SECTION 8

Key Findings & Strategic Recommendations

Synthesizing the analysis above into actionable conclusions.

```

findings = {
    "Market Leader (Revenue)": "Peloton at $717M – but declining YoY (-10%)",
    "Fastest Growing": "WHOOP at +18% YoY – strong wearable + data positioning",
    "Best Brand Equity": "Nike Training Club – massive reach, but weak monetization",
    "Most Vulnerable": "Lululemon Mirror – hardware pivot backfiring (-45% YoY)",
    "Highest User Satisfaction": "Peloton – 4.8 avg rating, loyal subscriber base",
    "Biggest Market Opportunity": "AI-personalized coaching + wearable integration",
}

recommendations = [
    "Freemium-to-premium funnel: The gap between Nike TC (free, high reach) and premium",
    "brands is an opportunity. A freemium model with a $30-40/mo premium upsell",
    "can capture price-sensitive users while building a path to high LTV.",
    "Content gap left by Peloton: Pelotons brand decline has opened a vacuum in",
    "aspirational guided fitness content. High-production content can build",
    "brand equity and organic search share without hardware investment.",
    "Data as the moat: WHOOPs success signals that biometric + behavioral data",
    "integration is the next retention battleground. Brands that build personalized",
    "feedback loops will win churn battles.",

```

```

"Community is non-negotiable: Pelotons most defensible asset is its community.",
"Any challenger must invest in social accountability features from day one.",
]

print("=" * 70)
print("KEY FINDINGS")
print("=" * 70)
for k, v in findings.items():
    print(f" {k}:\n {v}\n")

print("=" * 70)
print("STRATEGIC RECOMMENDATIONS")
print("=" * 70)
for i, r in enumerate(recommendations, 1):
    print(f" {i}. {r}\n")

```

— SECTION 9

Limitations & Next Steps

Limitations:

- Revenue figures for private companies (WHOOOP, Mirror) are analyst estimates
- Search interest data is simulated; real analysis would use the Google Trends API
- Positioning scores are qualitative and researcher-assigned, introducing subjectivity
- Analysis covers 2021–2024; the market continues to evolve rapidly

Potential next steps:

- Pull live Google Trends data via pytrends API for real search interest
- Scrape app store reviews with google-play-scraper for actual sentiment distribution
- Add social media follower growth and engagement rate analysis
- Build a regression model to predict revenue trajectory based on leading indicators

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